

Chapter 3: The Stress Hypothesis

Reading: [Fung/94] Chapter 3

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Contents

(§1.6) The Stress/Traction Hypothesis

Consider a material B occupying a spatial region V . The body is *transmitting* a balanced load system shown with *surface forces* in and *boundary constraints* in . A **traction field** arises within the body during this transmission. The word **traction** roughly describes the flow of load per unit area. To *illuminate* the traction field in a region *visualize* an enclosing surface S around it and *imagine* the interaction between the *inside* ($-$) and the *outside* ($+$) of S .

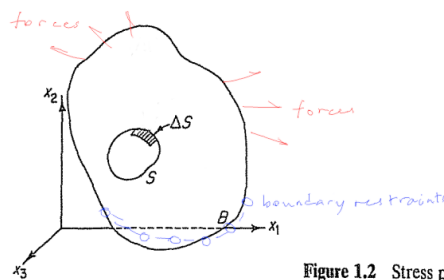


Figure 1.2 Stress principle.

The Stress/Traction Hypothesis

Let a neighborhood ΔS on S have an *outward unit normal* $\boldsymbol{\nu}$, with its direction pointing outward from the interior. Let the *accumulation of load* applied by the $(+)$ side on the $(-)$ side passing thru ΔS be represented by $\Delta \mathbf{F}$. The **stress hypothesis** can be stated: *as ΔS tends to a small but bounded size α , the ratio $\Delta \mathbf{F}/\Delta S$ tends to a definite limit $d\mathbf{F}/dS$, and the moment of the force acting on the surface ΔS about any point within the area*

vanishes. This limiting vector, called as the **traction** or the **stress vector** can be written as

$$\mathbf{T} = \mathbf{t} = \frac{d\mathbf{F}}{dS}.$$

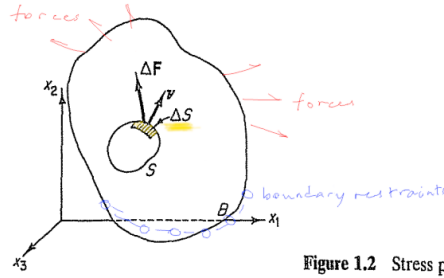


Figure 1.2 Stress principle.

Tractions on Cartesian Planes

- At an internal point, the standard procedure is to calculate the tractions $\mathbf{t}_1, \mathbf{t}_2, \mathbf{t}_3$ that arise on the planes cornered by a Cartesian triad $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$.

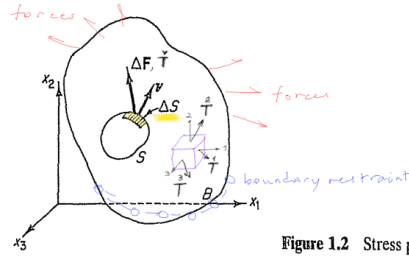


Figure 1.2 Stress principle.

- We combine the three Cartesian tractions as the columns of the transpose of a tensor $\boldsymbol{\tau}$ as

$$\begin{bmatrix} 1 & 2 & 3 \\ \mathbf{t}_1 & \mathbf{t}_2 & \mathbf{t}_3 \end{bmatrix} \triangleq [\boldsymbol{\tau}]^T$$

The Stress Tensor

- The tensor, $\boldsymbol{\tau}$, constructed by stacking the three Cartesian internal traction vectors in rows is called as the **stress tensor**. The stress

tensor components read

$$\begin{bmatrix} \tau_{11} & \tau_{12} & \tau_{13} \\ \tau_{21} & \tau_{22} & \tau_{23} \\ \tau_{31} & \tau_{32} & \tau_{33} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ t_1 & t_2 & t_3 \\ 2 & 2 & 2 \\ t_1 & t_2 & t_3 \\ 3 & 3 & 3 \\ t_1 & t_2 & t_3 \end{bmatrix} \quad \text{or} \quad \tau_{ij} = t_j^i$$

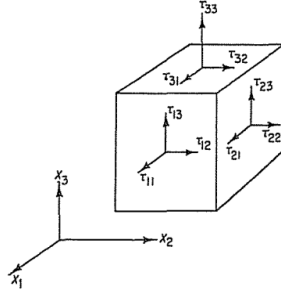


Figure 3.1 Notations of stress components.

Properties of Internal Traction Vectors

- The state of stress at a point in a material is described by internal traction vectors on a differential cube element.
- The differential cube has six faces, three of which have (+) normals and the other three have (−) normals.
- The three traction vectors corresponding to (+) faces are $\overset{1}{t}, \overset{2}{t}, \overset{3}{t}$, and the other three associated with the (−) faces are $\overset{-1}{t}, \overset{-2}{t}, \overset{-3}{t}$.
- The action-reaction principle of Newton can be used to relate the traction vectors on opposite cube faces as

$$\overset{i}{t} = -\overset{-i}{t} \quad \text{or} \quad \overset{i}{t}_j = -\overset{-i}{t}_j$$

- It is standard convention to decompose the traction vector, $\overset{\nu}{t}$, into its **normal** part, $(\nu \cdot \overset{\nu}{t})\nu$, and its remaining part, $\overset{\nu}{t} - (\nu \cdot \overset{\nu}{t})\nu$, which is **in-plane** of the surface.

Properties of Stress Components

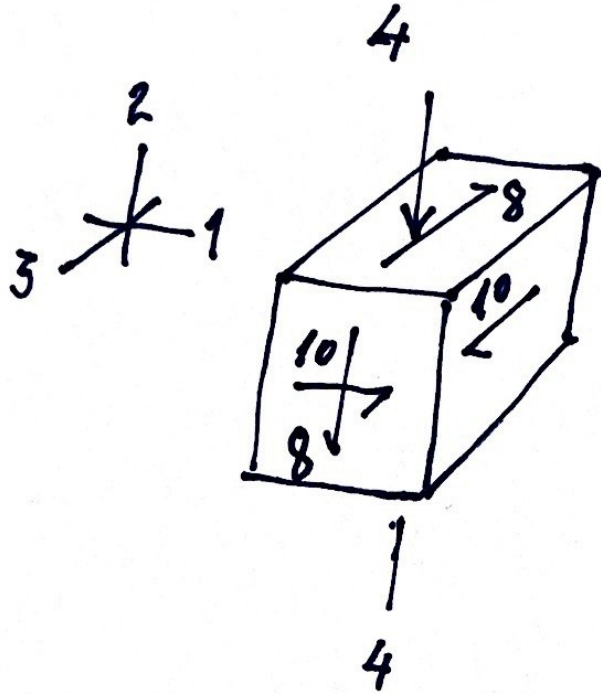
- In parallel with the parts of the internal traction vectors, the stress components that are normal to the plane are called as **normal stress** while those components which are in-plane are known as **shear stress**.

$$\text{normal} \begin{cases} \tau_{11} \\ \tau_{22} \\ \tau_{33} \end{cases} \quad \text{shear} \begin{cases} \tau_{12}, \tau_{13} \\ \tau_{21}, \tau_{23} \\ \tau_{31}, \tau_{32} \end{cases}$$

- The normal stresses are given a (+) sign if they point away from the surface (in **tension**) and a (−) sign if they point towards the surface (in **compression**).
- The shear component (e.g. τ_{ij}) is considered as positive if it is acting in the $+j$ direction on a plane that has a normal in $+i$ direction, or is in the $-j$ direction on a plane that has a normal in $-i$ direction.

Examples

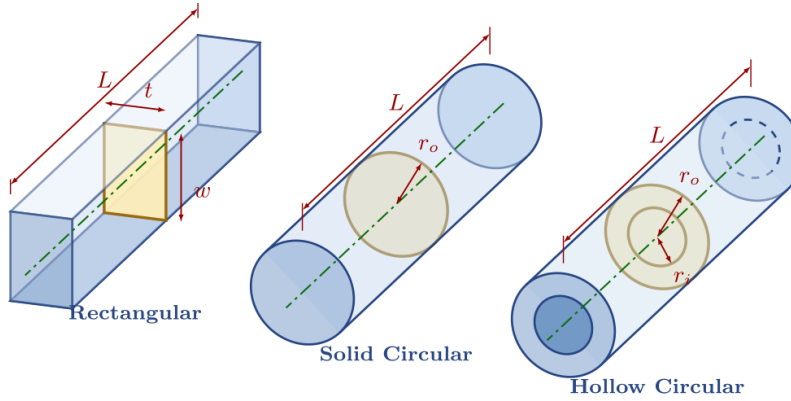
- Show the following stress components on a differential cube element $\tau_{11} = 12$ MPa, $\tau_{21} = -6$ MPa, $\tau_{31} = 5$ MPa, and $\tau_{33} = -8$ MPa, with all the remaining stress components identically equal to zero.
- Write the components shown on the stress cube in matrix form.



(§1.11) Examples to Elementary Traction Fields

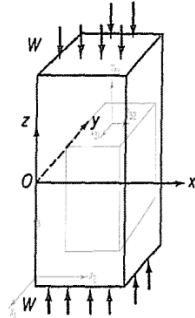
Slender Cylindrical (Axial) Members

A *member* (independent part of a structure) is said to be **slender**, if it has a great length to thickness ratio, i.e., $L/T \gg 1$, and a constant or slightly-varying cross section swept along the **normal** to the cross-section plane (a.k.a. member's *longitudinal* or *axial* direction). Such objects in geometry are known as *cylinders*.



Axially Loaded Cylindrical Members

- Consider a member under a **compressive** force W . We say that the beam is axially loaded without shear. Applying the stress hypothesis, we will obtain the relevant **traction vector**.



- Relabel coordinates ($x \rightarrow x_1, y \rightarrow x_2, z \rightarrow x_3$).
- Take a section of the member using the x_3 plane from the *mid-span* (region)

The mapping $\nu \rightarrow e_3$ yields $\overset{\nu}{t} \rightarrow \overset{3}{t}$

Derivation of Nominal Normal Stress (Assumptions and Conventions)

About the traction flow, we assume that:

- there is no reason that the load *would distribute non-uniformly* over the cross-section!! (i.e., there is **uniform distribution**, $dW/dA = W/A$, leading to

$$t_3^3 = \tau_{33} = \sigma = \frac{W}{A}.$$

Here σ denotes the **nominal normal** stress. It has units of load per area (e.g. N/cm²).

- Under uniaxial loading, and in planes with normals parallel to the load axis, **only normal traction** arises (no shear tractions, i.e. $t_1^3 = t_2^3 = 0$).
- Over planes that contain the load axis (whose normal will not have an e_3 component), **zero traction** prevails!! That means $t^1 = 0$ and $t^2 = 0$, which yield the state of stress

$$[\tau] = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \tau_{33} \end{bmatrix}$$

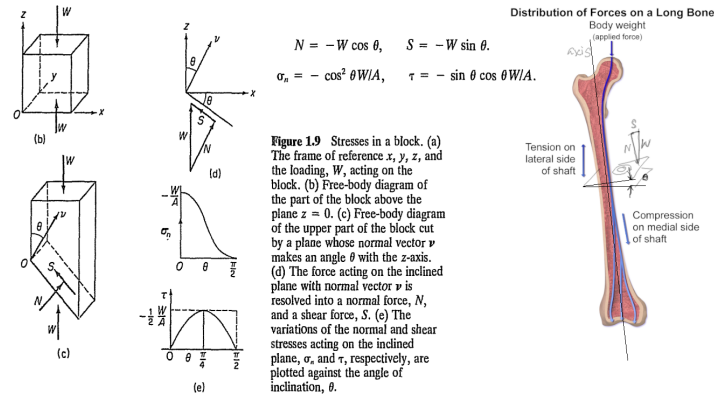
- The given state of stress (having all shear terms identically zero) is also known as a state of **principal stress**.

Example: Loading of the Femur

`./images/animation-femur.gif`

- Femur is the largest and most massive bone in the our body.
 1. Assume the load component in the axial-plane to be negligible?? (which is the same as saying the load is entirely axial and no shear component is taken)
 2. Using average values for a standard female, determine the normal compressive stress. (search keywords: CDC standard anthropometric measurements female data)
 3. Knowing the femur axis deviates from the vertical (gravity axis) on the average by some small angle (e.g. 9 degrees), what is a more realistic state of stress?

State of Stress on a Plane that is Not Normal to the Load.

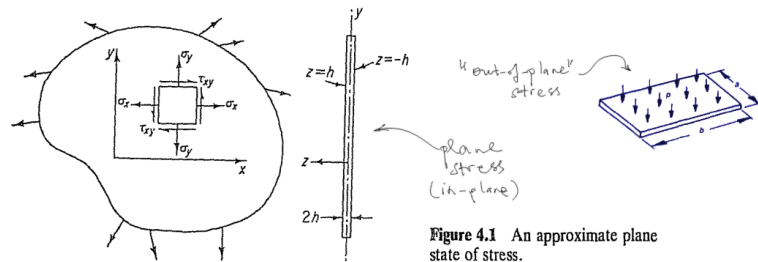


- Assume a deviation of 8 degrees between the axial load and the femoral axis, determine the shear stress to normal stress ratio observed on an axial plane.

Plates under Plane Stress

Plane State of Stress

- thin flat plates or membranes stretched by in-plane load systems



- found in ship hulls, pressurized bodies, wings, panels
- under these conditions the *out-of-plane* stress components vanish.
- if you choose a reference coordinate frame whose any two axes lie along the plate's plane, then you get a reduced stress tensor, such that (continue to next section)

Reduction of the Stress Tensor to 2-D

- We may choose the plate coordinate pair to be $(1, 2)$, $(2, 3)$, or $(1, 3)$, in which cases the corresponding stress tensors would read:

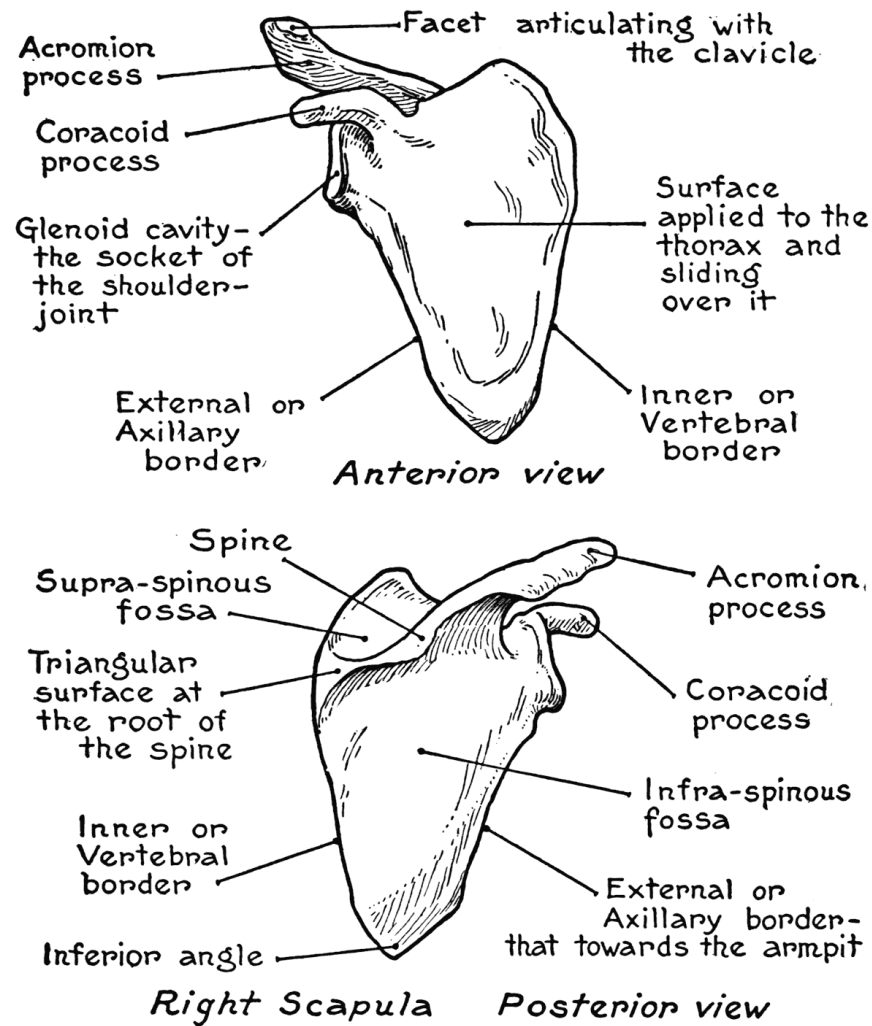
$$\tau_{ij} = [\tau] = \begin{bmatrix} \tau_{11} & \tau_{12} & 0 \\ \tau_{21} & \tau_{22} & 0 \\ 0 & 0 & 0 \end{bmatrix}, \{i, j = 1, 2\}, \quad \begin{bmatrix} 0 & 0 & 0 \\ 0 & \tau_{22} & \tau_{23} \\ 0 & \tau_{32} & \tau_{33} \end{bmatrix}, \{i, j = 2, 3\}, \quad \text{or}, \quad \begin{bmatrix} \tau_{11} & 0 & \tau_{13} \\ 0 & 0 & 0 \\ \tau_{31} & 0 & \tau_{33} \end{bmatrix}, \{i, j = 1, 3\}$$

- The two non-zero coordinate components can be combined in a *reduced* stress tensor, depicting the *state of plane stress* as

$$\tau_{ij} = [\tau] = \begin{bmatrix} \tau_{11} & \tau_{12} \\ \tau_{21} & \tau_{22} \end{bmatrix}, \{i, j = 1, 2\}$$

- There is no special symbol or indicator, whatsoever, that is used when the dimension of space is reduced from 3 to 2, $[\tau]$ prevails. The dimensionality is understood from the context.

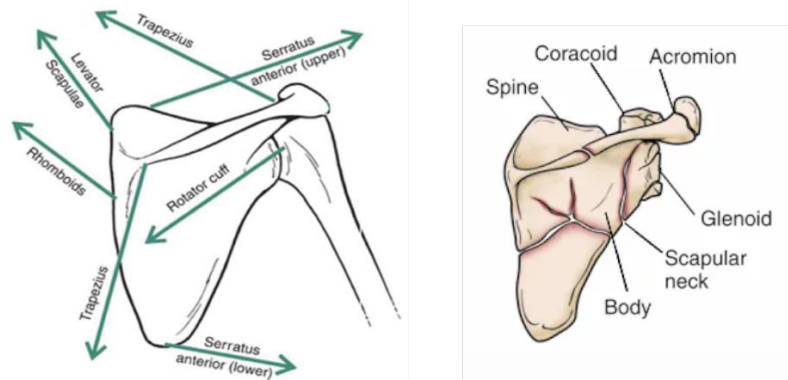
The Scapula (A flat bone)



THE SCAPULA AND ITS PARTS.

Loading and Fracture of Scapula

- The state of *in-plane* loading of the scapula, and the most frequently observed **fracture modes** (according to OrthoInfo)



- A nominal stress calculation does not exist for this case (details about loading and boundary restraints should be provided to pose a proper *boundary-value problem*)

Stresses in Pressurized Vessels

Many biological examples of pressurized biological tissue may be provided.

- Lung tissue: Maintains subatmospheric interstitial pressure (2 to 9 mmHg) due to elastic recoil, crucial for alveolar expansion and gas exchange.
- Kidney: Encapsulated organ with positive interstitial pressure (up to +10 mmHg), aiding fluid filtration and urine formation.
- Vasculatory system components, veins are pressurized biological tissues, but they operate under much lower pressure compared to arteries.

(§3.2) Laws of Motion

- Let the space occupied by a material body B at time t , be denoted it's configuration $B(t)$.
- Let $\mathbf{r}$ denote the position vector of a material point X .
- Utilize cartesian coordinates x_i , $i = 1, 2, 3$ for spatial measurements.
- Consider a differential volume element dv around X .
- Let ρ denote the mass density field and $\mathbf{v} = \dot{\mathbf{r}}$ denote the velocity field in B .

Balance of Mass

- The total mass of the material configuration $B(t)$ can be expressed by integration.
- We can denote the mass enclosed in dV by ρdV .
- Then the total mass of B at time t becomes

$$\mathcal{M} = \int_{B(t)} \rho dV$$

- The rate of change of the total mass is shown by

$$\dot{\mathcal{M}} = \frac{D\mathcal{M}}{Dt} = \frac{D}{Dt} \int_{B(t)} \rho dV = \int_{B(t)} \frac{D}{Dt} (\rho dV) = \int_{B(t)} (\rho dV)^{\bullet}$$

where $\frac{D}{Dt}(\dots)$ and $(\dots)^{\bullet}$ denote the *material time derivative* operator, taking the derivative with respect to configurational changes of the material body B with time.

- The conservation of mass of a body is expressed by setting

$$\dot{\mathcal{M}} = 0$$

- Notes of Importance

1. the balance of mass states that the net change in mass is given by the material time rate of the mass integral, $\dot{\mathcal{M}}$.
2. the role of material time derivative to highlight the difference between $\int_B(\dots)^{\bullet}$ and $\int_{B(t)}(\dots)^{\bullet}$
3. Ask GPT:

How would you describe the difference between a time derivative in the usual (*partial*) sense and that for a body in motion in the **total** sense? If possible, supply an image of a traveling (material) window versus a fixed (spatial) window highlighting the difference between tracking the motion of a group of particles versus observing the flow through a control volume.

Balance of Linear Momentum

- The total linear momentum of the material configuration $B(t)$ can also be expressed by integration.
- We denote the linear momentum of the mass ρdV by multiplying it with its velocity as $\mathbf{v}\rho dV$.
- Then the total linear momentum of B at time t becomes

$$\mathcal{P} = \int_{B(t)} \mathbf{v}\rho dV$$

- Euler interpreted Newton's equation of motion of a particle for a body by stating that the rate of change of linear momentum is equal to the resultant of all external mechanical effects (loads) as

$$\dot{\mathcal{P}} = \mathcal{F}$$

where $\dot{\mathcal{P}}$ denotes the material time derivative of the linear momentum and $\mathcal{F}$ denotes the net external force on B .

Classification of External Loads

- In the mechanics of continuous media, there are two types of external forces acting on material bodies.

1. Body forces, acting on volume elements (forces per unit volume).
 - $\mathbf{X}$ or $\mathbf{b}$
 - gravitational loads ($\rho\mathbf{g}$ has units of force per volume)

$$\int_{B(t)} \mathbf{b}\rho dV = \int_{B(t)} \mathbf{g}\rho dV$$

- electromagnetic loads

2. Surface forces, or surface tractions, acting on surface elements (forces per unit area).

- surface traction due to mechanical contact

$$\overset{\nu}{\mathbf{T}} \quad \text{or} \quad \overset{\nu}{\mathbf{t}} \quad \rightarrow \quad \int_{\partial B(t)} \overset{\nu}{\mathbf{t}} dA$$

- aerodynamic pressure (units of force per area), $-p\mathbf{n}$ where $\mathbf{n}$ is the unit outward normal at the surface of B .
- drag force in viscous flows

Balance of Linear Momentum (with forces)

Replacing the general expression of the resultant of external forces, $\mathcal{F}$ with the above force type definitions, we arrive at the terminal integral expression

$$\int_{\partial B(t)} \overset{\nu}{t} dA + \int_{B(t)} \mathbf{b} dV = \int_{B(t)} (\mathbf{v} \rho dV)^{\bullet}$$

Balance of Moment of Momentum

- The integral of the moment of momentum of the differential mass element about the origin O , $\mathbf{r} \times \mathbf{v} \rho dV$, over the domain $B(t)$, i.e.,

$$\mathcal{H}^O = \int_{B(t)} \mathbf{r} \times \mathbf{v} \rho dV$$

is the *moment of momentum* of the configuration at $B(t)$.

- According to Euler, the law of motion for a continuum asserts that the *rate of change of the moment of momentum is equal to the net applied torque $\mathbf{r} \times \mathcal{F}$ about the origin*, i.e.,

$$\dot{\mathcal{H}}^O = \mathbf{r} \times \mathcal{F}$$

- Replacing $\mathcal{F}$ with the force definitions, we the second vector equation of motion

$$\int_{\partial B(t)} \mathbf{r} \times \overset{\nu}{t} dA + \int_{B(t)} \mathbf{r} \times \mathbf{b} dV = \int_{B(t)} (\mathbf{r} \times \mathbf{v} \rho dV)^{\bullet}$$

(§3.3) Cauchy's Formula

Relates the components of the traction that resides over a plane with a normal ν , with the unit normal and the stress tensor components.

$$\overset{\nu}{t}_j = \nu_i \tau_{ij}$$

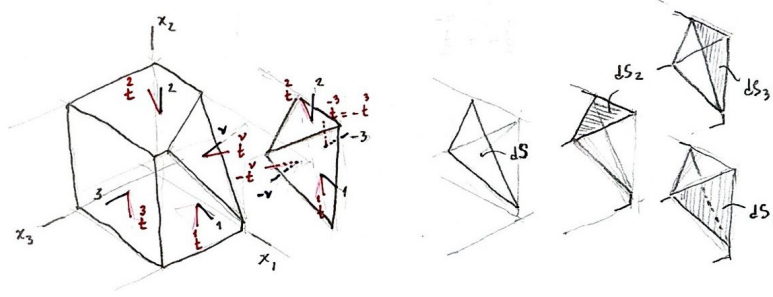
Note that similar relations between the traction vector components on cartesian planes, the base vector and stress tensor components exists:

$$\overset{1}{t}_j = \mathbf{e}_{1i} \tau_{ij} \tag{1}$$

$$\overset{2}{t}_j = \mathbf{e}_{2i} \tau_{ij} \tag{2}$$

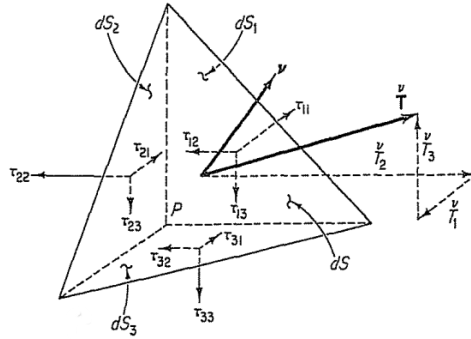
$$\overset{3}{t}_j = \mathbf{e}_{3i} \tau_{ij} \tag{3}$$

$$\tag{4}$$



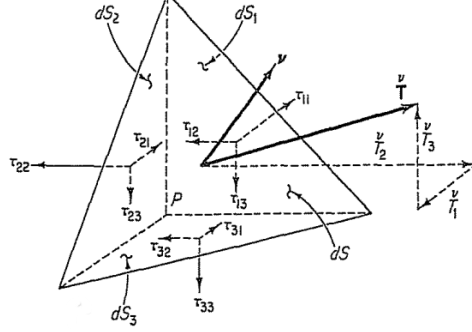
- The internal surface traction at a point P over a plane with a normal ν was defined by $\overset{\nu}{\mathbf{T}}$ or $\overset{\nu}{\mathbf{t}}$. The traction over the opposite plane (with a normal $-\nu$) is equal in magnitude but opposite in direction, i.e. $\overset{-\nu}{\mathbf{t}} = -\overset{\nu}{\mathbf{t}}$.
- This is also true for the tractions that lie over cartesian planes, e.g. $\overset{-1}{\mathbf{t}} = -\overset{1}{\mathbf{t}}$, etc.
- From geometry, we define the area formed by the intersection of an oblique plane with normal ν , and the three other cartesian planes, as dS .
- The projection of dS onto the cartesian plane i (perpendicular to x_i) is given by $dS_i = \nu_i dS$.

Motion Law Applied to a Tetrahedron Cut



- For the tetrahedron cut with four surfaces (dS_i, dS) and volume $dV = \frac{1}{3} h dS$:

- Surface forces: $-\overset{1}{t}dS_1, -\overset{2}{t}dS_2, -\overset{3}{t}dS_3$ on cartesian planes and $\overset{\nu}{t}dS$ on the oblique plane
- Body force: $\mathbf{b}\frac{1}{3}hdS$.
- Inertial term: $\dot{\mathbf{v}}\frac{1}{3}hdS$.

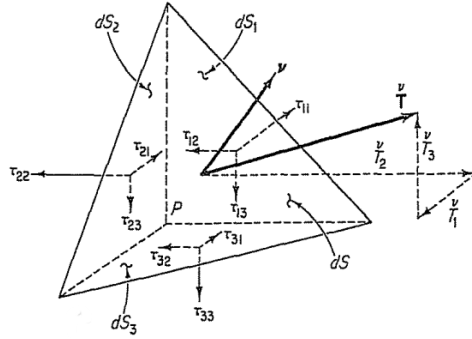


- The balance of linear momentum for the tetrahedron cut reads:

$$-\overset{1}{t}dS_1 - \overset{2}{t}dS_2 - \overset{3}{t}dS_3 + \overset{\nu}{t}dS + \mathbf{b}\frac{h}{3}dS = \dot{\mathbf{v}}\frac{h}{3}dS$$

- Replacing $dS_i = \nu_i dS$, dividing by dS , and taking the limit as $h \rightarrow 0$, we obtain

$$\overset{\nu}{t} = \nu_1 \overset{1}{t} + \nu_2 \overset{2}{t} + \nu_3 \overset{3}{t}$$



- Arranging the components ν_i in a vector form and the vectors $\overset{i}{t}$ in a matrix form yields

$$\overset{\nu}{t} = \{\nu_1 \ \nu_2 \ \nu_3\} \begin{bmatrix} \overset{1}{t} \\ \overset{2}{t} \\ \overset{3}{t} \end{bmatrix}$$

- and noting that the matrix components t_j^i corresponds identically with the stress tensor components τ_{ij} , we get the Cauchy's Formula $t_j^\nu = \nu_i \tau_{ij}$.

Sample Problem 1: (use plane formula)

3.7 The components of a stress tensor at a certain place in a body may be presented as a matrix:

$$\begin{matrix} & \begin{matrix} x & y & z \end{matrix} \\ \begin{matrix} x \\ y \\ z \end{matrix} & \begin{pmatrix} 0 & 1 & 2 \\ 1 & 2 & 0 \\ 2 & 0 & 1 \end{pmatrix} \end{matrix}.$$

What is the stress vector acting on the outer side (the side away from the origin) of the following plane, which passes through the place in question?

$$x + 3y + z = 1.$$

What are the normal and tangential components of the stress vector to this plane?

Answer: $\vec{T}_t = (5, 7, 3)/\sqrt{11}$; $T^a = \frac{29}{11} \cdot \text{shear} = 0.771$.

Sample Problem 2

3.8 With reference to the x-, y-, and z-coordinates, the state of stress at a certain point of a body is given by the following matrix:

$$(\sigma_{ij}) = \begin{pmatrix} 200 & 400 & 300 \\ 400 & 0 & 0 \\ 300 & 0 & -100 \end{pmatrix} \text{ kPa}$$

Find the stress vector acting on a plane passing through the point and parallel to the plane $x + 2y + 2z - 6 = 0$.

Answer: $\vec{T} = 533\mathbf{i} + 133\mathbf{j} + 33\mathbf{k}$.

(§3.5) Transformation of Stress Components

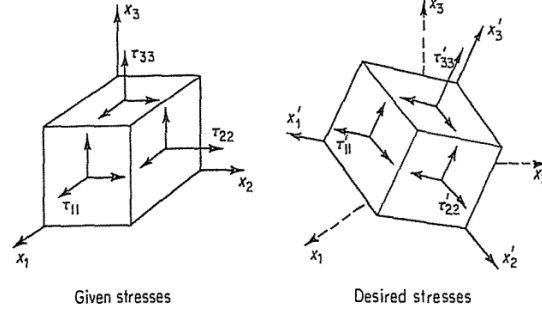
Remember how coordinates of a point, x_j , could be transformed from the basis $\mathbf{e}_j$ into another bases $\mathbf{e}_{i'}$?

- We used the rotation tensor definition $\beta_{i'j} = \mathbf{e}_{i'} \cdot \mathbf{e}_j$ for this purpose, and wrote the transformed coordinates as

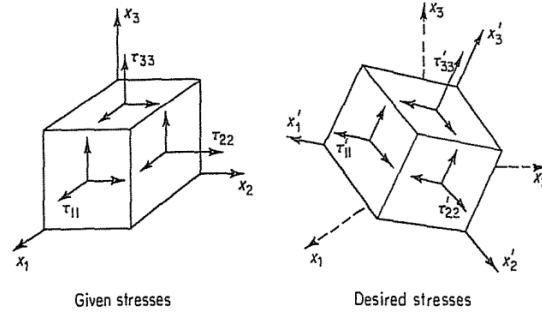
$$x_{i'} = \beta_{i'j} x_j$$

- Can this approach be used to transform the stress tensor components, $\tau_{ij} \rightarrow \tau_{i'j'}$?

The Rotation Tensor Transforms a Second-Order Tensor



- Stress is a second-order tensor with two indices τ_{ij} where the components are expressed with respect to a particular coordinate frame using a set of basis vectors, $\mathbf{e}_i$.
- Consider an alternative coordinate frame using it's own set of basis vectors, $\mathbf{e}_{j'}$.
- How are the stress components in the two frames, τ_{ij} and $\tau_{i'j'}$ related?



- As it turns out each one of the indices of a higher-order tensor can be and should be transformed by a coordinate transformation operation
- Thus, for a second-order tensor such as stress, two transformations are in order, as

$$\tau_{i'j'} = \beta_{i'i} \beta_{j'j} \tau_{ij}.$$

- Noting the order in which inner product takes place, the matrix form reveals

$$[\tau'] = [\beta][\tau][\beta]^T$$

where a normal matrix multiplication occurs in a row-column operation which a transposed multiplication occurs in a column-column operation.

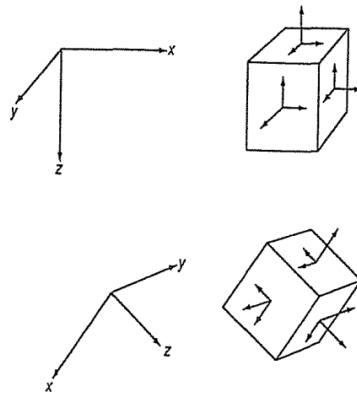
Educational Application Hub:

Tools for transforming components of the second-order stress tensor

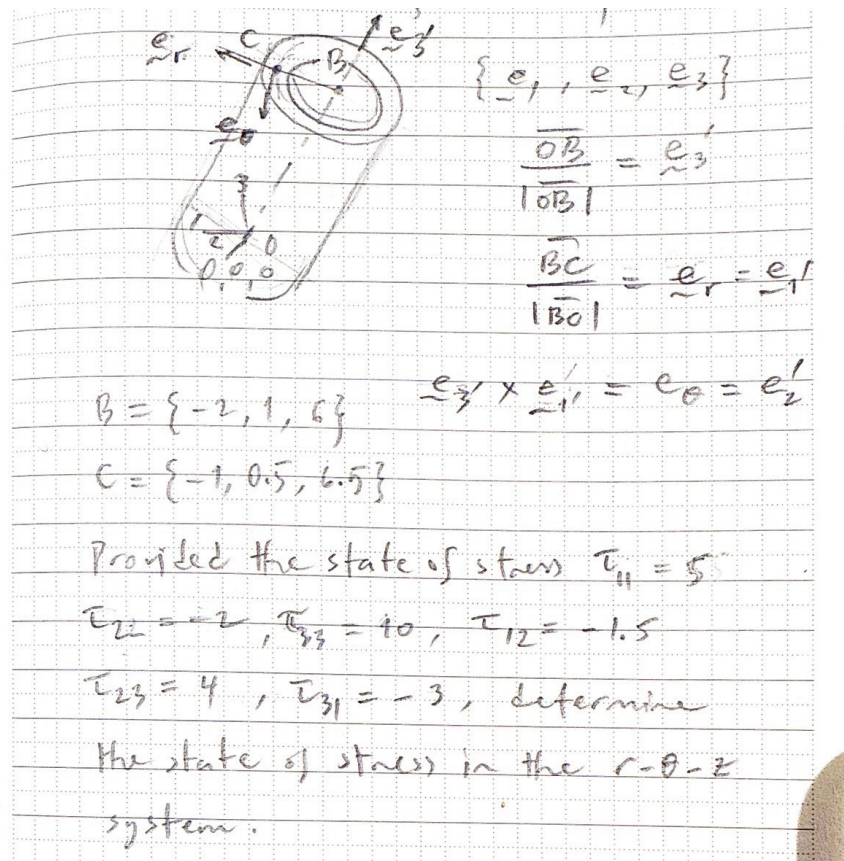
- <https://eduapphub.com/apps/app.php?dir=2Dstresstransform>
- <https://eduapphub.com/apps/app.php?dir=3Dstresstransform>

Sample Problem 1

3.6 Label the stresses shown in Fig. P3.6.



Sample Problem 2



- Initial script produced in class

```
B = [-2 1 6] % point definition, upper end of axis
C = [-1 0.5 6.5] % point definition, state of stress is given here
BC = C - B % line segment from axis (is this perpendicular to OB?)
e_1p = BC / norm(BC) % e_r ??
e_3p = [0 0 1] % e_z ??
e_2p = cross(e_3p, e_1p)
beta = [e_1p; e_2p; e_3p]
```

- Two problems with this script can be identified
 - the cylinder axis may not be aligned with x_3
 - Segment BC may not be purely radial

- Corrected script

```

e_BC = BC / norm(BC)    % vector may not be radial, do not call it e_1p
e_3p = B / norm(B)      % vector e_3p is not [0 0 1], OB -> B
C_3p = dot(C,e_3p)      % scalar projection of C onto e_3p
CT3p = C - C_3p*e_3p    % vector component of C perpendicular to e_3p
e_1p = CT3p / norm(CT3p)
e_2p = cross(e_3p,e_1p)
beta=[e_1p;e_2p;e_3p]

```

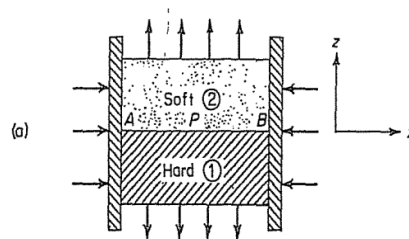
(§3.7) Boundary Conditions

Describing various state of stress conditions that arise in bodies.

- condition of a body submerged in a fluid
 - a building subjected to wind
 - a vessel wall subjected to internal pressure
- condition at a bimaterial interface
 - relative sliding and contact between bodies
 - sticking interface (various joining methods exist)
- condition of point loads (singularities)
 - point forces and force couples

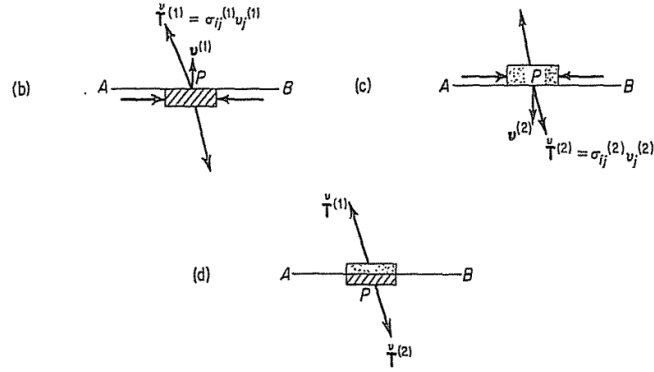
We know something about the forces or motion on the surface of a solid or a fluid and inquire what happens within ...

A Bimaterial Square Block Under Axial Contraction



- Imagine the square is made of a soft upper half and a hard lower part joined together by the middle horizontal line.

- Let the square block be compressed by the lateral side walls on the left and right. What will be the state of stress at the bimaterial interface?
- How will the state of stress change within each body?



- From the interface, traction vector changes slightly within the soft part (a), and the hard part (b)
- At the interface, boundary condition requires the *continuity of traction*,
i.e. $t_i^{(1)} = t_i^{(2)}$.

What is external?

Although every surface is an interface between two spaces, it is a general practice to confine one's attention to one side of the surface and call the other side "external." For example, structural engineers speak of the wind load on a building as the "external load" applied to the building. Reciprocally, to the fluid dynamicist, the building is merely a rigid border to the flow of air. The same interface presents two different kinds of boundary conditions to the two media. The basic justification for such a divergence of attitude is that the small elastic deformation of the structure is unimportant to the aerodynamicist who computes the aerodynamic pressure acting on the structure, whereas the elastic deformation is all important to the structural analyst who determines the safety of the building. Hence, for the aerodynamicist the building is rigid, whereas for the elastician it is not. In other words, both boundary conditions are approximations.

Sample Problem: Hydrostatic Stress

3.4 In Fig. P3.4, water is shown in a reservoir. At a point P , let us consider surfaces $A-A$, $B-B$, etc. Draw stress vectors acting on these surfaces. Consider all possible surfaces passing through P . What is the locus of all the stress vectors?

Answer: A sphere.

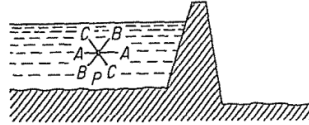


Figure P3.4 Water in a reservoir.